

• Regresión Lineal (Regression)

Modelo

target $y \in \mathbb{R}$
numérico

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_m x_m = X\beta$$

$m \rightarrow \# \text{ features}$

$$\begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1m} \\ 1 & x_{21} & x_{22} & \dots & x_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{nm} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{pmatrix}$$

Sol.
Analítico

$$\beta = (X^T X)^{-1} X^T Y$$

Función de Costo / Error

$$\mathcal{J}(\vec{\beta}) = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

$n \rightarrow \# \text{ datos}$

Encuentra $\vec{\beta}$ que minimizan

$\text{argmin}(\mathcal{J})$

$$\nabla = \left(\frac{\partial}{\partial \beta_0}, \frac{\partial}{\partial \beta_1}, \dots \right)$$

Descenso del Gradiente

$$\beta_{k+1} = \beta_k - \underbrace{\alpha}_{\text{learning-rate}} \nabla \mathcal{J}$$

• Regresión Logística (Clasificación) y $\rightarrow$ $\{0, 1\}$ Binario
 $\{1, 2, 3, \dots\}$ Multiclases

• Modelo

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_m x_m$$

$$= X\beta \quad (-\infty, \infty)$$

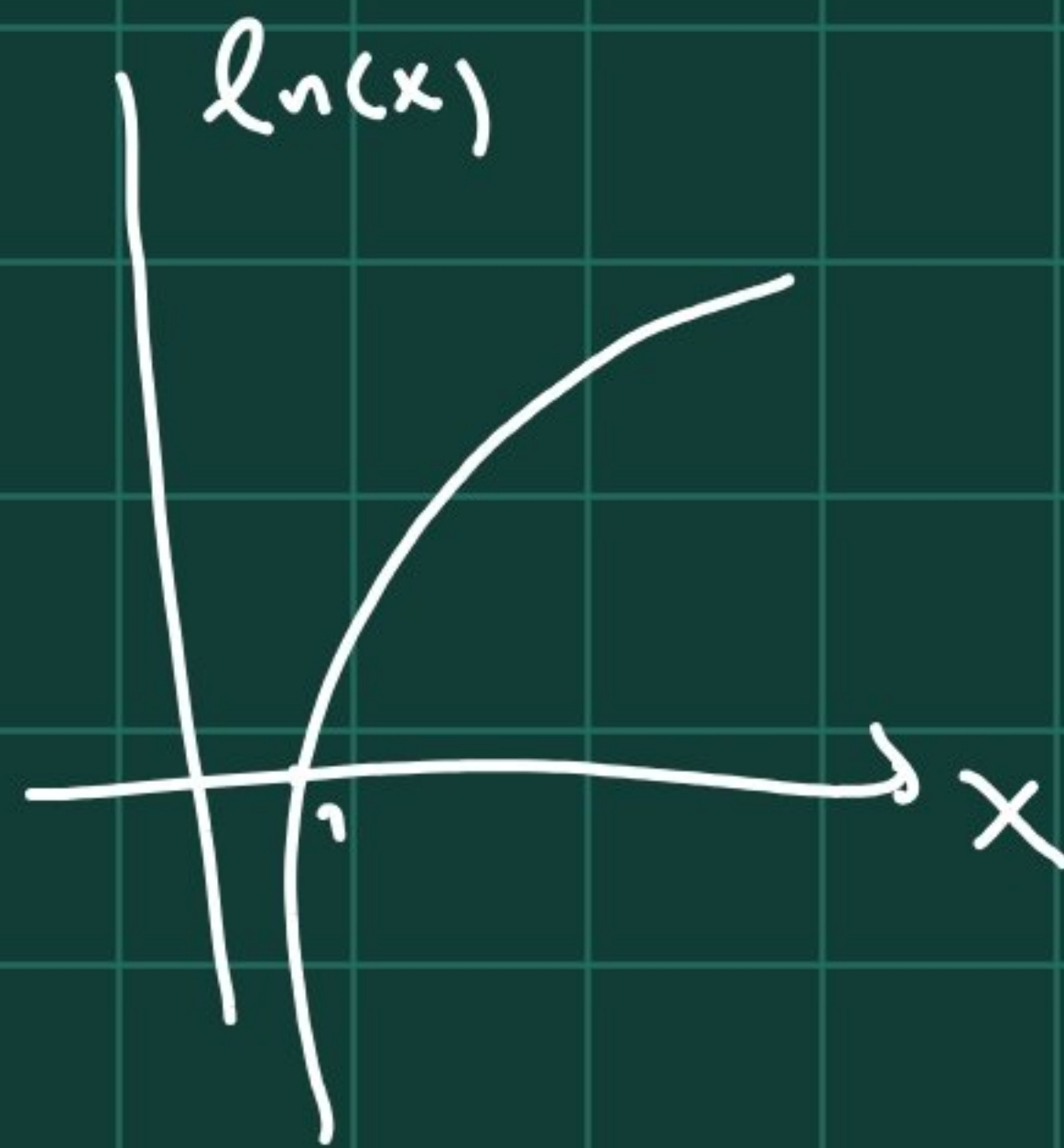
$\hat{y}$
 $\{0, 1\}$
 x

P
 $[0, 1]$

$\frac{P}{1-P}$
 $(0, \infty)$
 ODDS

$\ln\left(\frac{P}{1-P}\right)$
 $(-\infty, \infty)$
 LOG-ODDS

$\rightarrow \frac{\text{éxito}}{\text{fracaso}}$



Modelo $\ln\left(\frac{P}{1-P}\right) = X\beta$
log-odds.

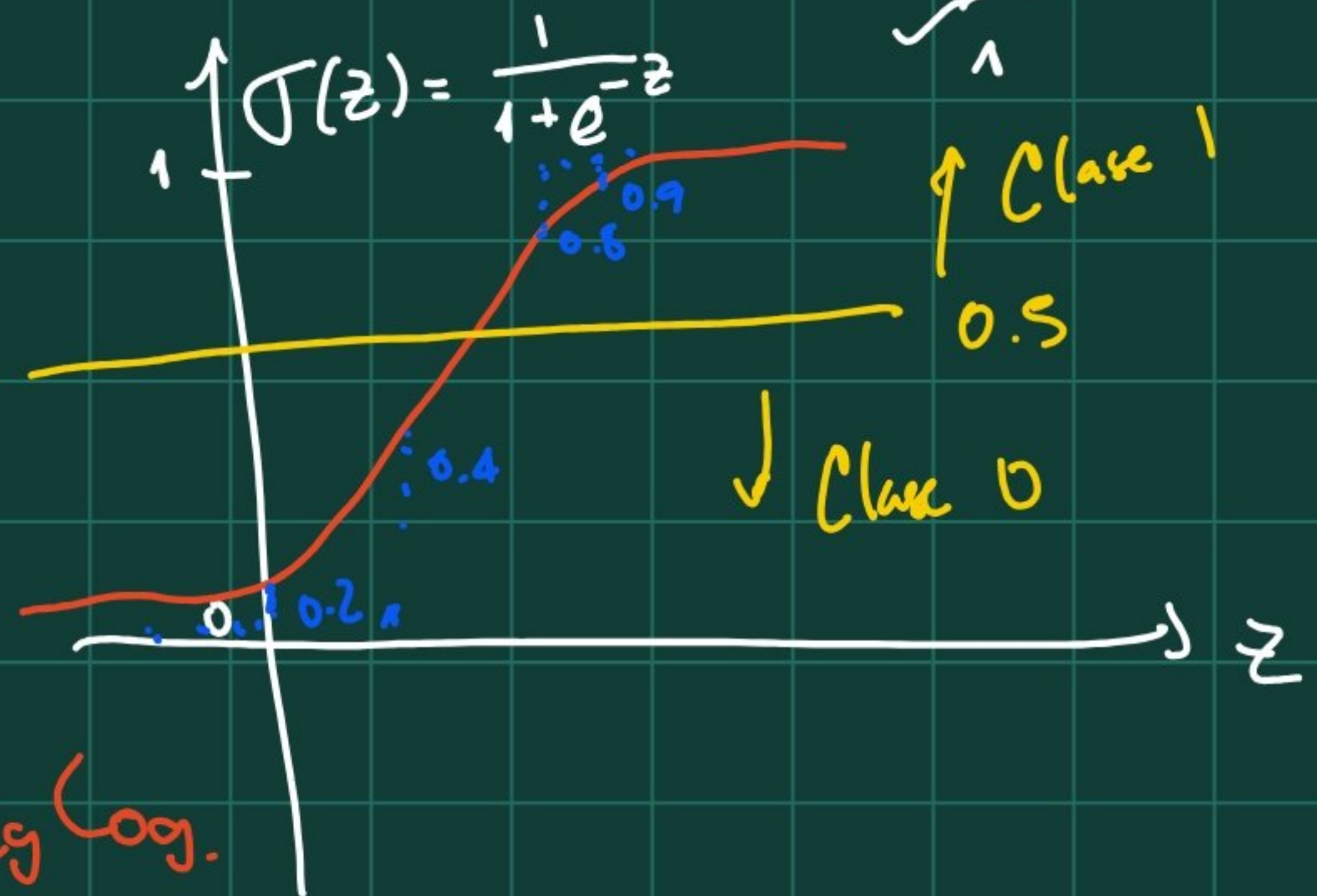
$$\exp\left(\ln\left(\frac{P}{1-P}\right)\right) = \exp(X\beta) \Rightarrow \frac{P}{1-P} = e^{X\beta} \Rightarrow P = (1-P)e^{X\beta} = e^{X\beta} - Pe^{X\beta}$$

$$\Rightarrow P + Pe^{X\beta} = e^{X\beta} \Rightarrow P(1 + e^{X\beta}) = e^{X\beta} \Rightarrow P = \frac{e^{X\beta}}{1 + e^{X\beta}} \underbrace{\left(\frac{e^{-X\beta}}{e^{-X\beta}}\right)}_1$$

$$\Rightarrow P = \frac{1}{1 + e^{-X\beta}}$$

(0,1)

Función Sigmoid



! Esto es lo que predice la Reg Log.

• Modelo

$$I = \frac{1}{1 + e^{x\beta}}$$

$$\Rightarrow I = \sigma(x\beta)$$

Sigmoide $\Rightarrow \sigma(z) = \frac{1}{1 + e^{-z}}$

$$\begin{aligned} 1 - \sigma(z) &= 1 - \frac{1}{1 + e^{-z}} \\ &= \frac{(1 + e^{-z}) - 1}{1 + e^{-z}} = \frac{e^{-z}}{1 + e^{-z}} \end{aligned}$$

Propiedad

$$\frac{d\sigma(z)}{dz} = \frac{d}{dz} \left(\frac{1}{1 + e^{-z}} \right) = \frac{d}{dz} \underbrace{(1 + e^{-z})^{-1}}_u$$

Regla de la cadena

$$\frac{d\sigma}{dz} = \frac{d\sigma}{du} \frac{du}{dz}$$

$$= \left(-1 u^{-2} \right) \left(-e^{-z} \right) = \left(1 + e^{-z} \right)^{-2} (e^{-z}) = \frac{e^{-z}}{(1 + e^{-z})^2} = \left(\frac{1}{1 + e^{-z}} \right) \left(\frac{e^{-z}}{1 + e^{-z}} \right)$$

$\sigma \qquad 1 - \sigma$

$$\therefore \boxed{\sigma'(z) = \sigma(z)(1 - \sigma(z))}$$

- Función de Cost

D. Bernoulli:

$$P(X=1) = p$$

$$P(X=0) = 1-p$$

Parámetro
 p

$$\begin{aligned} P(X_1 \cap X_2 \cap X_3) \\ = P(X_1) \cdot P(X_2) \cdot P(X_3) \end{aligned}$$

Ej. Supongamos que tenemos 6 características independientes
 $X_1, X_2, X_3, \dots, X_6$.

¿Cuál es la prob. de observación?

$$X_1 = X_2 = X_3 = 1, \quad X_4 = X_5 = X_6 = 0$$

Suponiendo que siguen D. Bernoulli:

$$P(X_1 = X_2 = X_3 = 1, X_4 = X_5 = X_6 = 0 \mid p)$$

$$= P(X_1=1 \mid p) \cdot P(X_2=1 \mid p) \cdot \dots \cdot P(X_6=0 \mid p)$$

$$= p \cdot p \cdot p \cdot (1-p) \cdot (1-p) \cdot (1-p)$$

$$= p^3 (1-p)^3$$

DATOS			
x_1^{age}	x_2^{estud}	x_3^{sueto}	y pasó?
18	8	6	1
20	2	1	1
28	6	8	0

$$P(\bar{x}_1, \bar{x}_2, \bar{x}_3 | P(\beta))$$

$$= P_1 \cdot P_2 \cdot (1 - P_3)$$

$$P(\beta) = \frac{1}{1 + e^{-x\beta}}$$

Función de Verosimilitud

$$y_i \rightarrow \{0, 1\}$$

$$L = \prod_{i=1}^n P_i^{y_i} \cdot (1-P_i)^{1-y_i}$$

param's
 β 's

$$P^{(i)} = \frac{1}{1 + e^{-x^{(i)}\beta}}$$

$i \rightarrow \text{filn}$

$$\sum x_i = x_1 + x_2 + \dots$$

$$\prod x_i = x_1 \cdot x_2 \cdot \dots$$

Objetivo: $\text{argmax}(L)$

Método de Máxima Verosimilitud

Regla del prod $\frac{d(f(x)g(x))}{dx} = f'(x)g(x) + f(x)g'(x)$

$$\ln L = \ell = \sum_{i=1}^n y_i \ln(P_i) + (1-y_i) \ln(1-P_i)$$

Optimización

Log. Propiedades

$$\log(ab) = \log(a) + \log(b)$$

$$\log(x^a) = a \log(x)$$

Función de Costo/Erro $\rightarrow$ Entropía Cruzada Binaria
(BCE)

- Función de Costo / Error

$$l = - \sum y^{(i)} \ln(p^{(i)}) + (1 - y^{(i)}) \ln(1 - p^{(i)})$$

$$\operatorname{argmax}(l) \rightarrow \operatorname{argmin}(-l)$$

Optimizar

Recordamos

$$p = \frac{1}{1 + e^{-x\beta}} = \sigma(x\beta); \quad \frac{d\sigma}{dz} = \sigma(1 - \sigma)$$

$$\frac{\partial l}{\partial \beta_j} = - \sum \frac{\partial}{\partial \beta_j} \left(y^{(i)} \ln(\sigma^{(i)}(x\beta)) \right) + \frac{\partial}{\partial \beta_j} \left((1 - y^{(i)}) \ln(1 - \sigma^{(i)}(x\beta)) \right)$$

$\beta_0, \beta_1, \beta_2, \dots$

$$x\beta = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots = \beta_j x_j^{(i)} \text{ cols}$$

$$\frac{d}{dx} (\ln(x)) = \frac{1}{x}$$

$$= - \sum y^{(i)} \frac{1}{\sigma(x\beta)} \frac{\partial}{\partial \beta_j} \sigma(x\beta) + (1 - y^{(i)}) \frac{1}{1 - \sigma(x\beta)} \frac{\partial}{\partial \beta_j} (-\sigma(x\beta))$$

$$\frac{\partial}{\partial \beta_j} \left(\sigma(\underbrace{x\beta}_u) \right) = \frac{\partial \sigma}{\partial u} \cdot \frac{\partial u}{\partial \beta_j}$$

$$x\beta = \beta_0 + \beta_1 x_1 + \dots + \beta_m x_m$$

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots \\ 1 & x_{21} & x_{22} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

$j = 0 \quad 1 \quad 2 \quad 3 \dots$

$$= \sigma(x\beta) (1 - \sigma(x\beta)) x_j$$

$$\begin{aligned}
\frac{\partial \ell}{\partial \beta_j} &= - \sum y^{(i)} \frac{1}{\sigma(x\beta)} \frac{\partial}{\partial \beta_j} \sigma(x\beta) + (1-y^{(i)}) \frac{1}{1-\sigma(x\beta)} \frac{\partial}{\partial \beta_j} (-\sigma(x\beta)) \\
&= - \sum_{i=0}^n y^{(i)} \frac{1}{\cancel{\sigma(x\beta)}} (\cancel{\sigma(x\beta)} (1-\sigma(x\beta)) x_j^{(i)} + (1-y^{(i)}) \frac{1}{1-\cancel{\sigma(x\beta)}} (-\cancel{\sigma(x\beta)} (1-\cancel{\sigma(x\beta)})) x_j^{(i)} \\
&= - \sum_{i=0}^n [y^{(i)} (1-P^{(i)}) + (1-y^{(i)}) (-P^{(i)})] x_j^{(i)} \\
&= - \sum_{i=0}^n [y^{(i)} - \cancel{y^{(i)} P^{(i)}} + (-P^{(i)}) + \cancel{y^{(i)} P^{(i)}}] x_j^{(i)}
\end{aligned}$$

$$\frac{\partial \ell}{\partial \beta_j} = - \sum_{i=0}^n [\underbrace{y^{(i)}}_u - \underbrace{P^{(i)}}_v] \underbrace{x_j^{(i)}}_v = 0 \quad \text{! No hay solución analítica} \Rightarrow \text{aproximados}$$

cond. de minimización

$$\underbrace{X^T (y - P)}_{\frac{\partial \ell}{\partial \beta_j}} \rightarrow \nabla \ell$$

$$\sum u_i v_i = u_1 v_1 + u_2 v_2 + u_3 v_3 + \dots$$

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}; \quad P = \begin{pmatrix} P_1 \\ P_2 \\ \vdots \\ P_n \end{pmatrix}; \quad X = \begin{pmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \ddots & \ddots \\ 1 & \ddots & x_{nn} \end{pmatrix}$$

$$\boxed{\beta_{k+1} = \beta_k - \alpha \nabla \ell}$$

$$\frac{\partial \ell}{\partial \beta_j} = - \sum_{i=0}^n [y^{(i)} - P^{(i)}] x_j^{(i)} \quad \text{Primer derivada.}$$

$$P^{(i)} = \sigma(x^{(i)} \beta)$$

> Segunda Derivada (Matriz Hessiana)

$$\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k} = \begin{pmatrix} \frac{\partial^2 \ell}{\partial \beta_0 \partial \beta_0} & \frac{\partial^2 \ell}{\partial \beta_0 \partial \beta_1} & \cdots \\ \frac{\partial^2 \ell}{\partial \beta_1 \partial \beta_0} & \frac{\partial^2 \ell}{\partial \beta_1 \partial \beta_1} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

$$\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k} = - \frac{\partial}{\partial \beta_k} \sum (y^{(i)} - \sigma(x^{(i)} \beta)) x_j^{(i)}$$

$$= - \sum \frac{\partial}{\partial \beta_k} (y^{(i)} - \sigma(x^{(i)} \beta)) x_j^{(i)} = \sum \frac{\partial}{\partial \beta_k} (\sigma(x^{(i)} \beta)) x_j^{(i)}$$

$$= \sum_{i=1}^n x_k^{(i)} (\sigma(x^{(i)} \beta)) (1 - \sigma(x^{(i)} \beta)) x_j^{(i)} = \sum_{i=1}^n x_k^{(i)} (P^{(i)}) (1 - P^{(i)}) x_j^{(i)}$$

$$\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k} = \sum_{i=1}^n \underbrace{x_k^{(i)} (p^{(i)})(1-p^{(i)})}_{W^{(i)}} x_j^{(i)} = \sum_{i=1}^n x_k^{(i)} W^{(i)} x_j^{(i)}$$

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}; \quad W = \begin{pmatrix} p_1(1-p_1) & 0 & 0 & 0 & \vdots \\ 0 & p_2(1-p_2) & 0 & 0 & \vdots \\ 0 & 0 & p_3(1-p_3) & 0 & \vdots \\ 0 & 0 & 0 & p_4(1-p_4) & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

$$\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k} = X^T W X$$

Matrix
Hessian